

Policies have prose reviews, but no unit tests

A prototype for turning session-confirmed travel & expense intent into traceable scenarios and deterministic checks.

SYNTHETIC PROTOTYPE · VALIDATION PENDING

Narrow claim

**Make ambiguity
executable — then test it.**

One owner carries every edge case in their head

FINANCE / PEOPLE OPS

Policy owner

Maintains rules, fields exceptions, and explains decisions across many combinations of amount, category, timing, destination, and prior spend.

HYPOTHESIS — NOT USER RESEARCH

1

Thresholds hide boundary gaps

Manual review can miss equality cases and adjacent ranges.

2

Exceptions collide across clauses

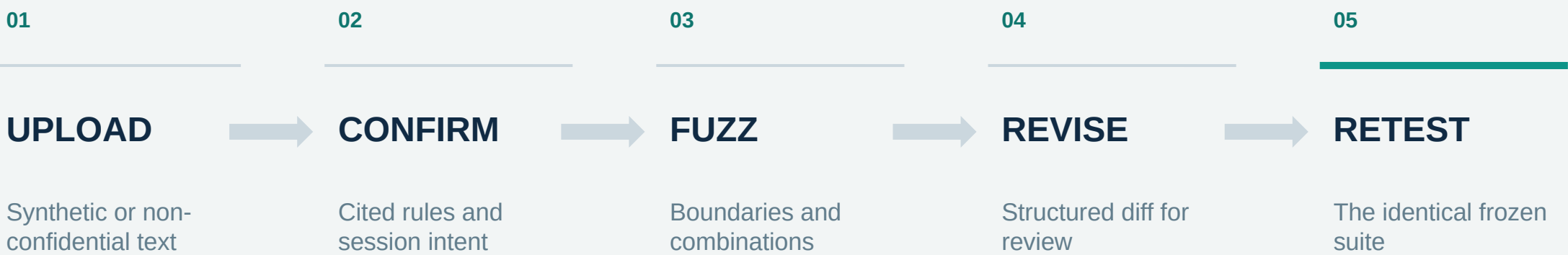
Priority and overrides may exist only in reviewers' memory.

3

Change review lacks a fixed regression set

A wording fix can shift a different outcome without a visible trace.

The journey ends in a retest, not a summary

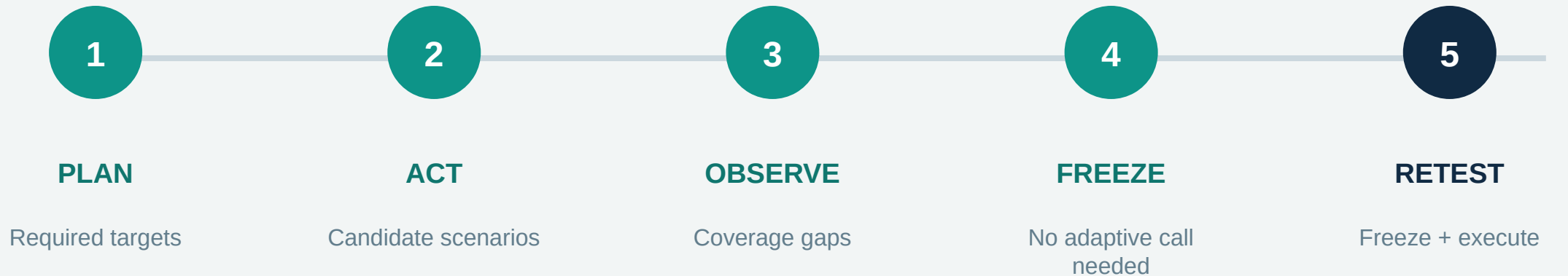


Why it differs from summarization

Every finding should resolve to visible facts, a deterministic trace, and an exact source citation — then survive the same-suite comparison.

LIVE / CACHED MODE STAYS VISIBLE

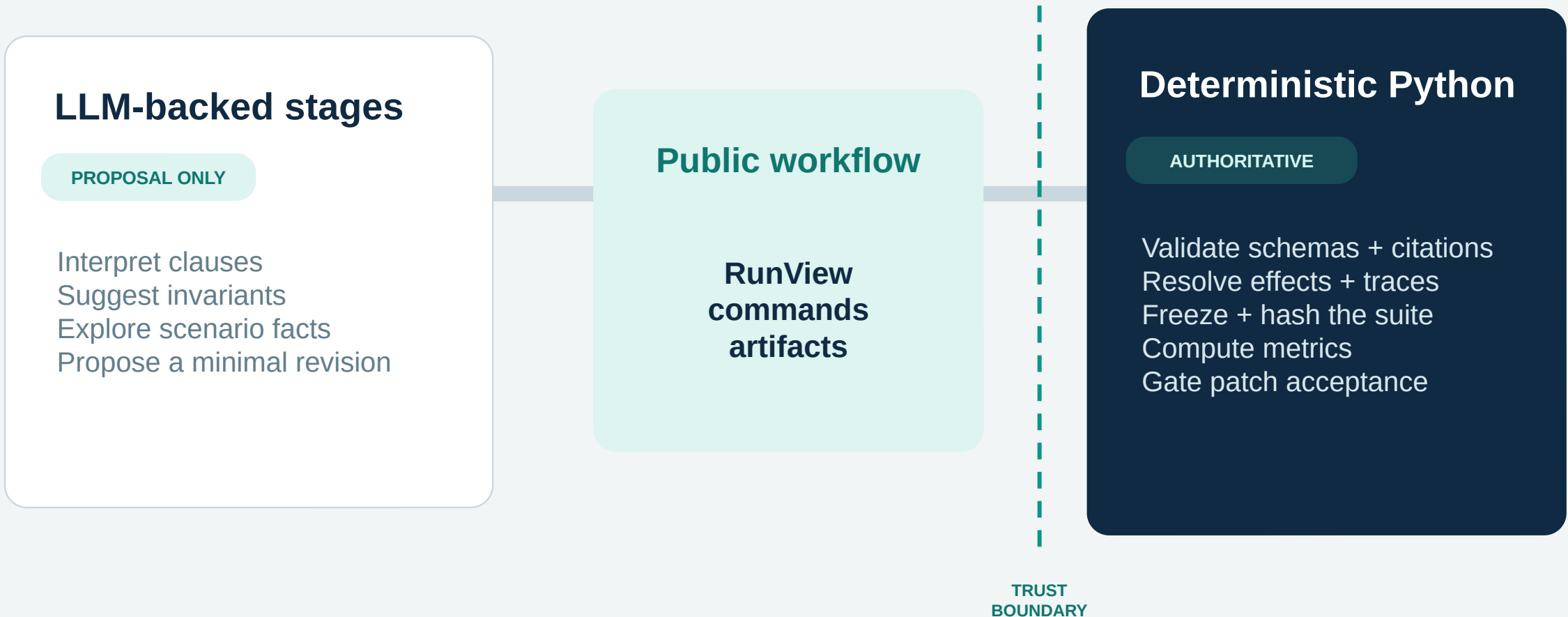
Coverage satisfied on the first bounded pass



Observed

0 targeted cycles · minimum coverage already satisfied.

Models propose; deterministic Python decides



Three deterministic findings make ambiguity executable

01	RECEIPT REQUIREMENT Deterministic baseline finding	Visible witness + cited rules Accepted by authored demo decision	FROZEN CAPTURE PENDING
02	APPROVAL REQUIREMENT Deterministic baseline finding	Visible witness + cited rules Accepted by authored demo decision	FROZEN CAPTURE PENDING
03	DAILY CATEGORY CAP Deterministic baseline finding	Visible witness + cited intent Accepted by authored demo decision	FROZEN CAPTURE PENDING

CACHED Current record: 3 baseline findings. Final witnesses, traces, and citations still require the frozen browser capture.

3 findings become 0; all seven gates pass

NOT FROZEN · NOT BLIND-ELIGIBLE

BEFORE

3 findings

10 scenarios
12 inconclusive assertions
11 gaps · 1 conflict
Minimum coverage satisfied

AFTER

0 findings

Same frozen suite
12 passing assertions
0 gaps · 0 conflicts
7 / 7 gates true

PROVENANCE

Cached rehearsal

4 scripted model responses
0 actual provider calls
Synthetic data
Blind-scoring-ineligible

Final chain **Gate B** → **demo-core-v1** → **6-hour freeze** → **browser capture** → **two human reviews**

Until that chain completes, these numbers remain rehearsal evidence — not headline blind results.

The impact hypothesis is better review evidence

TODAY

Policy review relies on memory, spot checks, and examples that change between reviewers.

NO ROI OR SAVINGS CLAIM BEFORE MEASUREMENT

Future proof
points

time to first actionable finding · total run time · confirmations · model calls · provider cost or N/A

HYPOTHESIS

1

Broader review coverage

Systematic boundaries and combinations expose what was exercised.

2

Traceable decisions

Facts, rule paths, and citations make disagreements inspectable.

3

Regression evidence

The same frozen suite tests a confirmed structured change.

A narrow prototype with a clear finish line

The intended proof

A session-confirmed structured change fixes targeted failures and introduces no protected regressions in the disclosed frozen suite.

VALIDATION PENDING

Next: Gate B, 6-hour freeze, browser capture, two human reviews.

Limits

Prototype; validation pending
Synthetic T&E scope
No legal-compliance claim
No automatic publication
Human confirmation remains required

Five-person ownership

P1 integration · P2 policy · P3 fuzzing
P4 evaluation · P5 product

github.com/Luigi-Simon/policyfuzz